

ERNOS LABS · SMITHIAN FOLD THEORY

UNISONAI

A computational proof of derived law with zero trained parameters

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LIVE DERIVED-LAW ARCHITECTURE

277,583,049 observations; 212,395,127 canonical entries

One-to-one attention, reward-conditioned learning, and real-engagement generalisation.

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Abstract

UnisonAI is the artificial-intelligence computational proof of the Smithian Fold Theory of Everything (SFT). SFT begins with one machine-checked, self-proven theorem—*there is no nothing*—with zero axioms; the theorem forces the One and fold. Unison asks whether the operations normally purchased through trained parameters can instead be forced, forward-forced, or constitutionally re-derived as inspectable laws.

The current architecture has zero trained parameters in its native memory, binding, generation, context, perception, learning, and verification mechanisms. The main SFT corpus executes 326 proof suites and 2,002 checks with zero failures. Its Unison laws include Step 322 coherence (8/8), Step 323 contextual integration (14/14), and Step 324 generation selection (13/13). The current standalone omni verifier passes 21/21 checks; the prior complete native-transformer suite receipt is 55/55 and the new v5 runtime's focused checks pass 4/4. The sealed v4 causal store contains 649,917 role-bound pairs and 11,140,970 assistant targets; the live v5 position relation contains 277,583,049 observations and 212,395,127 unique canonical entries. These are executable receipts for the derived-law architecture. Benchmark victory and continued forward forcing are active objectives; agents do not declare walls or endpoints.

This release separates theorem-forced quantities, constitutionally re-derived algorithms, and disclosed engineering quantities. That provenance distinction protects the argument: engineering constants are the current implementation status of a component, not a claim that the component is permanently outside SFT.

1. The derived-law thesis

Contemporary AI usually acquires structure through parameter optimisation. Unison tests a different route: derive the required computational law, execute it transparently, and halt if its declared invariants fail. The relevant comparison is therefore not “statistics versus hand-written heuristics.” It is trained statistical acquisition versus machine-checked derived law.

Unison is one of four SFT computational proofs, alongside Protein, Chess, and Go. Operational performance is an active development and benchmark-victory surface. It does not replace the engine's standard for forcing validity. A development batch remains a measurement of its named build until Maria declares the project finding and next direction.

2. Foundation and provenance

Every promoted mechanism must enter through one of three routes:

1. **Direct forcing:** derived from the theorem and earlier closed SFT steps.
2. **Forward forcing:** a new SFT derivation that closes a required computational operation.
3. **Constitutional re-derivation:** an established operation reconstructed under SFT mathematics and constraints.

Current engineering choices are labelled rather than hidden. The BM25 constants used by the pair-retrieval instrument are one example: they are neither trained parameters nor represented as Step 322's theorem-forced coherence value 1/2. The former word/generic fallback and its sub-half serving thresholds were agent-authored interpretations, not derived Unison components; that path is retired rather than treated as an unfinished SFT derivation.

3. Architecture

3.1 Held memory

Experiences are stored as deterministically addressed exact counts. The memory surface is inspectable and can be corrected without changing an opaque weight field. Exact rational next-token shares remain available as a measured substrate.

3.2 Binding and selection

The native conversational route binds complete prompt tokens and positions through counted Q/K compatibility and unit-capacity attention. Role-structured pair retrieval remains a separately disclosed RAG/response-selection instrument; it is not the native generator or a fallback. Prompt-admitted induction can transfer held token relations, while the full-statement-copy guard prevents a stored answer from being emitted wholesale as if recall were generation.

3.3 Context and generation

Step 323 forces the five-stage dyadic context cascade and closes it to the One. Step 324 forces the half-One spread/binding lock and the binary minimum of held expressions for fresh re-expression. Conversational generalisation is being built as a one-to-one constitutional translation of established causal attention-transformer and training computation into the SFT model; it is not a reconstruction of the retired fallback. Agent-selected development measurements are retained with their ownership; they are not declared as Maria's findings or losses.

The sealed native v4 artifact is the current executable translation. One exact role-bound pass over 649,917 prompt/assistant pairs deposited 11,140,970 causal targets, 22,415,744 counted Q/K relations, 96,721 key-owned value vectors, and 81,111,826 deep contextual FFN addresses. Every prompt token—including function words, punctuation, and repetition—enters attention. Structural, inverse-exposure information, and conditional counted-association heads close separately before exact residual addition and LM-head normalization. The 2,921,639,096-byte artifact is sealed at SHA-256
f977d4d8adb0993a0ab0d63b86dea30bd845ae091e84a4859ae826d032a87219.

An exact serving translation now stores all seven complete count tables in memory-mapped index/data files with no sampling, pruning, quantization, fitted capacity, or row eviction. Its receipt binds every packed-file hash to the sealed v4 artifact and runtime source; manifest or source drift halts. Exact integer accumulation and a response-local bounded row cache preserve the materialized rational head's token choice. On the fixed eight-prompt Codex probe, resident memory fell from approximately 59 GB to 559,185,920 bytes and generation time fell from 1,514.370707 seconds to 11.407350 seconds while every surface remained byte-identical across the representation change.

Distinct prompt occurrences now retain turn age, within-turn position, and sequence position through five exact contextual attention/FFN blocks. Counted embedding Q/K products select the complete theorem-traced dyadic cascade; the closing remainder stays on the identity address, shared-denominator integer state preserves exact rational output, and the final prompt position supplies the decoder readout. The complete hidden support is the observed prompt position set, with no cap or dense corpus-layer materialization.

On the fixed Codex eight-prompt probe, all contextual states closed exactly; the eight promoted-route generations took 19.627877 seconds with 701,612,032 bytes maximum resident memory. Seven surfaces were unchanged and public speaking changed. This is an implementation observation, not Maria's benchmark result. The final contextual position basis now remains distinct through decoder value and semantic-FFN routing; exact tests prove those branches recombine to the sealed v4 token-owned rows.

That position-conditioned training port is now complete. One canonical (relative position, prompt token, last, previous, next) -> count relation preserves all **277,583,049** observations once and supplies position-owned values, semantic FFN2, and semantic FFN3 by exact marginalisation. It contains **212,395,127** unique entries in **4,369,646,672 bytes** and is sealed at SHA-256

99f1e032a600a8b3625d5266658323b3f0a0bb0c542b4ee674877b838ca093b0. The exact packed representation contains **5,097,483,048 bytes** and is sealed at SHA-256

f1e355cfb0e93d781d79452e12c58bdb3e95635acb5408f7a98deb9000ca579c. No observation, position, or corpus row is sampled, pruned, capped, fitted, or triplicated.

The packed v5 relation is now integrated into the live autoregressive runtime, sealed together with the unchanged v4 counted store and the exact runtime sources. For every retained prompt occurrence, the decoder reads exact relative-position value, semantic2, and semantic3 projections from the same canonical relation. There is no historical word/generic fallback and an unobserved deeper address contributes no invented row. On the fixed eight-prompt applied panel, all **628/628** value queries were observed; semantic2 supplied 577 observed addresses and semantic3 supplied 476. The panel completed in **66.154159 seconds** with **678,100,992 bytes** maximum

resident memory and preserves every prompt, surface, runtime address census, artifact identity, and source hash. This is applied implementation evidence; Maria Smith determines its conclusion and benchmark status.

Applied native evidence also exercises contextual transfer. Role-conditioned induction/copy attention produced **4/4** continuity on the matched multi-turn panel and **8/8** exact continuations on a separate transfer panel. Requiring the prompt itself to admit the induction relation preserved both measurements, removed every full-statement copy, and produced non-empty output on all four free follow-ups. These are implementation measurements of the native route; Maria Smith determines the conclusion and benchmark timing.

3.4 Perception and common memory

Sight and hearing use integer Walsh transforms and certify Parseval identity for each act. Their signatures enter the same held-memory constitution as language. Speech can replay a held record without a synthesis model; new synthesis remains a separate organ.

3.5 Learning and observers

Corrections, counted quality, and teacher comparisons are written to explicit ledgers. Supporting models are scaffolding, not Unison's native parameters. The architecture's intended removal condition is a counted graduation lock; current empirical graduation is reported only where the relevant ledger supports it.

4. Machine-checked evidence

4.1 Main SFT corpus

Surface	Receipt
Current proof suites	326
Current forced checks	2,002
Failures	0
Source/certificate identities	326/326
Step 322 coherence	8/8
Step 323 contextual integration	14/14
Step 324 generation selection	13/13

4.2 Standalone Unison

The prior complete Python repository-suite receipt passes **55/55**, including tests for the native transformer role boundary, complete prompt-token retention, active Q/K weighting, exact distribution closure, lazy/full argmax equivalence, packed-row and surface identity, bounded-cache identity, sealed-source drift halts, native live routing, and reward-ledger separation. The standalone python3 omni/benchmarks/verify_omni.py receipt remains the separate **21/21** structural instrument covering forced locks, halt-on-mismatch, sight, hearing, held memory, exact shares, diagnostics, removal-proof voice, and persistence across process death. The integrated v5 path adds focused canonical-relation, runtime-score, activation, and legacy-closure checks, **4/4** passing in the targeted new-code execution.

The older verify_unison_results.txt is a historical **43/47** receipt. Earlier prose incorrectly called it 47/47; this release corrects that documentary drift rather than blending the historical and current suites.

4.3 Implemented prediction, learning, and conversation evidence

The counted character engine measured **1.2891** against the matched trained twin's **1.8878** on the same recorded split and metric. The counted word engine measured **3.1907** against **3.4292**. The source-backed taught loop rose from **17% to 75%** after one teaching round. In the native transformer, matched induction continuity reached **4/4**, separate exact transfer reached **8/8**, and prompt admission retained those results while removing full-statement copy.

These are positive implemented results for prediction, learning, context, and transfer. Full free conversational composition and benchmark victory remain the active forward surfaces. Agent-authored auxiliary benchmark rows

are preserved in train_eval/REGISTERED_GENERATION_QUALITY.md with their ownership and protocol; they are not imported here as Maria Smith's findings, losses, parity definition, or evidence of a theoretical limit.

5. What zero parameters means here

“Zero trained parameters” means the native Unison architecture contains no learned weight tensor and performs no gradient optimisation to acquire its laws. It does not mean that operating-system limits, file formats, buffer sizes, model-scaffolding settings, or every current engineering constant are theorem quantities. Those categories are explicitly separated.

The stronger objective—every steering quantity forced or constitutionally re-derived—remains active. Recording current engineering quantities is a development ledger for that objective, not a retreat from it.

6. Relation to the Fold Decode campaign

The companion paper, *The Law Inside Trained Weights*, studies trained models as measured objects. It reports registered spectral structure, matched nulls, ablation evidence, and training chronology. Agent-authored expectations are identified separately from the measurements. Those measurements motivate and test the derived-law thesis, but they are not silently imported as Unison's mechanism.

7. Reproduction

```
# Current standalone structural receipt
python3 omni/benchmarks/verify_omni.py

# Main-corpus current-source and certificate receipt
cd "/path/to/Smithian Fold Theory"
./verify/prove_current_source_isolated.sh
```

The decode instruments and their raw receipts are documented in omni/benchmarks/INTERPRETABILITY.md and omni/benchmarks/REPRODUCE.md.

8. Current stage, next state, and theoretical reach

The complete v5 position relation is now active in native serving. Its exact packed identity, unchanged counted-store identity, position-owned projections, and applied eight-prompt address census are sealed. The next work is exact serving optimisation and source-bound held-out free multi-turn conversation, with reward-conditioned learning preserved. The same real path then advances through Maria-authorized benchmark runs.

Full conversational generalisation is not theoretically walled off by the current evidence. The established causal-transformer organs already execute in SFT form: role-bound training, complete tokens and positions, five contextual blocks, multi-head Q/K/V, semantic FFNs, residual, normalization to the One, LM head, autoregression, reward-conditioned learning, and exact packed serving. Counted prediction has beaten matched trained twins; contextual continuity and transfer have executed; and the complete uncapped 277.6-million-observation position relation is active. The current frontier is continued applied conversation, learning, and benchmark development, not a missing theorem-derived principle. Preserve every measurement with its provenance. Agents do not declare development runs to be Maria's findings, losses, or endpoint.

Repositories and lineage

- UnisonAI: <https://github.com/MettaMazza/UnisonAI>
- Main SFT corpus: <https://github.com/MettaMazza/Smithian-Fold-Theory-Of-Everything>
- Architecture concept DOI: <https://doi.org/10.5281/zenodo.21217278>
- Companion decode concept DOI: <https://doi.org/10.5281/zenodo.21396742>